

Modus Technical Architecture & Executive Overview

1. Core Technology Stack

Modus is built on a high-performance, scalable, and secure architecture designed for the creator economy.

- **Frontend: React Native (Expo) with TypeScript.**
 - *Why:* Allows for cross-platform (iOS/Android) parity while maintaining a single, type-safe codebase. TypeScript ensures architectural stability as the project scales.
 - **Backend & Database: Supabase (PostgreSQL).**
 - *Role:* Handles relational data, real-time subscriptions, and enterprise-grade security.
 - **Authentication: Supabase Auth (including Google OAuth).**
 - *Role:* Provides frictionless onboarding for brands and creators with secure session management.
 - **Storage: Supabase Storage.**
 - *Role:* Securely hosts high-resolution deliverables and creator portfolios.
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2. Key Architectural Components

A. The Campaign Builder (Wizard Engine)

We implemented a multi-step, state-driven "Wizard" for campaign creation.

- **Technical Implementation:** Uses a progressive disclosure pattern to minimize cognitive load. State is managed locally during creation and synced to PostgreSQL only upon finalization.
- **Visibility Logic:** A custom visibility attribute ('private' | 'public') was implemented at the core of the campaign schema to enforce strict data privacy.

B. "Manage Candidates" Hub (Dual-Channel Recruitment)

A sophisticated candidate management system that distinguishes between inbound applications and outbound invites.

- **Real-time Data Joins:** Complex SQL queries link campaign_offers with profiles to provide real-time status updates (e.g., "Awaiting Creator Response").
- **UX Distinction:** Features a tabbed architecture that dynamically hides or shows recruitment channels based on the campaign's privacy settings.

C. Creator "Opportunities" Board (Public Marketplace)

A discovery feed for creators to find and apply to projects.

- **Security Filtering:** Implements database-level filtering to ensure "Private" campaigns never leak

into the public discovery layer.

- **The "No Free Work" Protocol:** The pitch modal is restricted to text-only inputs, preventing brands from requesting unpaid creative assets before hiring.

D. Vault Escrow & Asset Management (Security Layer)

A digital safe for campaign funds and finalized content.

- **Status Machine:** Orchestrates the transition of a campaign from draft -> active (funded) -> submitted -> completed.
 - **Digital Receipting:** Assets in the "Vault" are linked to usage rights and license agreements stored in the database.
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3. Security & Scalability

- **Row Level Security (RLS):** We implemented strict RLS policies in PostgreSQL. This ensures that a creator can only see the campaigns they are invited to or that are public, and brands can only access their own campaign data.
 - **Normalized Schema:** The database is structured to support multi-creator campaigns (One Campaign -> Many Offers/Applications), allowing brands to hire and manage teams at scale.
 - **Optimistic UI Updates:** The app uses immediate UI feedback loops for actions like deleting campaigns or submitting pitches, with background synchronization to ensure a snappy user experience.
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4. Visual & UI Logic

- **Design Tokens:** Custom design system using HSL color scales and premium typography (Inter/System).
 - **Iconography:** Integration of **Lucide React Native** for high-fidelity, consistent visual cues.
 - **Responsive Layouts:** Utilizing `useSafeAreaInsets` and `Dimensions` to ensure a premium look on everything from iPhone SE to the latest Android flagships.
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Modus is not just an app; it's a secure, scalable operating system for brand-creator collaborations.